STANDATA bulletin 23-BCB-002

Building

Roof truss layout for one- and two-family dwellings

Date Issued: August 2024

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Purpose

To provide information regarding the status for the acceptance of roof truss layout drawings for one- and two-family dwellings that typically would not require professional involvement unless the scope of the construction or component is not addressed by Part 9 of the National Building Code – 2023 Alberta Edition (NBC(AE)).

Discussion

STANDATA 19-BCI-023 for Manufactured Engineered Components (archived) was published in April 2021 and provided specific circumstances where manufactured engineering components need not be designed by a registered engineering professional – all other circumstances required the placement diagrams and design drawings to be authenticated by a registered engineering professional. The interpretation defined roof trusses as an included item in the definition of manufactured engineering components. This interpretation allowed builders to submit a supplier's letter, design drawings, and placement diagrams to be provided as part of the building permit application as well as the manufacturer's corporate letter when placement diagrams and design drawings were generated by proprietary design software.

STANDATA 19-BCB-009 was published in December 2022 and withdraw the above referenced interpretation as a result of unintended consequences revealed by municipalities, industry and the Association of Professional Engineers and Geoscientists of Alberta (APEGA) regarding:

- the requirement for engineer authentication on truss layouts and profiles,
- the capacity of Alberta's safety code system to meet the requirements, and
- the scope of application for professional involvement related to the process for roof truss layouts.

The bulletin noted the authority having jurisdiction may accept industry practices at that time to continue affording industry additional time to develop a process that would comply with all applicable legislation and governing bodies.

The National Building Code – 2023 Alberta Edition (NBC(AE)) came into force on May 1, 2024 and confirmed requirements in 9.23.14.11 for roof trusses to be designed in accordance with good engineering practices and for joint connections to be designed in accordance with Part 4. The associated note recognized that Part 9 requirements do not usually require involvement by a registered professional, however noted that wood roof trusses are complex structures.

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Removing STANDATA 19-BCI-23 created a gap in understanding for industry and governing bodies around the role or scope of professional involvement and authentication of roof trusses. While larger structures require professional involvement in accordance with the requirements of Part 4, one- and two-family dwellings typically would not require professional involvement unless the scope of the construction or component is not addressed by Part 9 of the NBC(AE).

Until clarity is determined regarding the identified unintended consequences noted above, the involvement of a registered professional is not a minimum requirement for wood roof truss layouts or profiles and therefore engineering stamps or seals are not required as was previously the case in Alberta for one- and two-family dwellings constructed in compliance with Part 9 of the NBC(AE).

Note: This Standata only addresses wood roof trusses for the construction of one- and two-family dwellings.

Safety codes officers continue to have the authority under Section 18 of the Administrative Items Regulation (a regulation under the *Safety Codes Act*) to require all plans and specifications, or any part of them, be imprinted with a seal or stamp affixed in accordance with the *Engineering and Geoscience Professions Act* for engineering work, or the *Architects Act* for architectural work, if in the opinion of the safety codes officer, the size or complexity of a project raises safety concerns.

Under Section 18, a safety codes officer may also require that any or all of the project construction be reviewed during construction in accordance with the *Engineering and Geoscience Professions Act* or the *Architects Act* if, in the opinion of the safety codes officer, the size or complexity of a project raises safety concerns.

Code References

Sentence 2.4.1.1.(5) Division C

2.4.1. Professional Involvement
2.4.1.1. General
5) Drawings and specifications required to be authenticated by a *registered professional* shall be authenticated by the respective professional in accordance with the requirements of the Architects Act or the Engineering and Geoscience Professions Act.

Sentence 2.4.1.2.(1) Division C

2.4.1.2. Structural Design

1) For design carried out in accordance with Part 4 of Division B, the *designer* shall be a *registered professional* skilled in the *work* concerned. (See Note A-2.4.1.2.(1).)

A-2.4.1.2.(1) Structural Design. Part 4 of Division B is written on the assumption that structural design will be carried out by a registered professional who is qualified to perform such design. Sentence 2.4.1.2.(1) is not intended to imply that a registered professional may not also be required in the application of requirements in other Parts of the NBC(AE).

Sentences 9.23.14.11. (1) and (2)

9.23.14.11. Roof Trusses
1) Wood roof trusses shall be designed in accordance with good engineering practice such as that described in TPIC 2019, “Truss Design Procedures and Specifications for Light Metal Plate Connected Wood Trusses.”
2) The joint connections used in trusses described in Sentence (1) shall be designed in conformance with the requirements in Subsection 4.3.1. (See Note A-9.23.14.11.(2).)

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A-9.23.14.11.(2) Wood Roof Truss Connections. Sentence 9.23.14.11.(2) requires that the connections used in wood roof trusses be designed in conformance with Subsection 4.3.1., and Sentence 2.4.1.2.(1) of Division C, which applies to all of Part 4, requires that the designer be a registered professional skilled in the work concerned. This has the effect of requiring that the trusses themselves be designed by registered professionals. Although this is a departure from the usual practice in Part 9, it is appropriate, since wood roof trusses are complex structures which depend on a number of components (chord members, web members, cross-bracing, connectors) working together to function safely. This complexity precludes the standardization of truss design into tables comprehensive enough to satisfy the variety of roof designs required by the housing industry.

Illustrated Users Guide NBC 2015: Part 9 of Division B

9.23.14.11. Roof Trusses
This Article indicates the requirements for roof trusses to ensure that they will be capable of supporting anticipated roof and ceiling loads without collapsing, and without excessive deflections that may damage the ceiling finish or create objectionable sagging. Roof trusses have largely replaced joists and rafters in roofs.

Trusses are available in a wide variety of geometries and are generally constructed to the designs developed by truss plate manufacturers or truss fabricators.

Allowable deflections for roof trusses are provided in NBC Table 9.23.14.11. It is important that trusses be supported at their support points exactly the way they were intended in the design. Trusses must be supported where the design specifies such support. Field modifications must not be made unless approved by the designer. Any alteration to the original design, including support location, cutting, or notching is not permitted unless it can be shown that the alteration is safe.

Roof trusses are very strong in their vertical position provided the top members are supported against buckling by the roof sheathing. In their horizontal position, however, they are very weak and can easily be damaged. Trusses should, therefore, always be lifted in their vertical position and transported vertically, unless they are fully supported.

NBC Article 9.23.14.11. requires that the connections used in wood roof trusses be designed in conformance with NBC Part 4. NBC Sentence 4.1.1.5.(2), which applies to all of NBC Part 4, requires that a designer be a specially trained and qualified person skilled in the work concerned. This has the effect of requiring that the trusses themselves be designed by professional engineers or architects. Although this is a departure from the usual practice in the NBC, it is appropriate, since wood roof trusses are complex structures that depend on a number of components (e.g., chord members, web members, cross-bracing, connectors) working together to function safely. This complexity precludes the standardization of truss design into tables comprehensive enough to satisfy the variety of roof designs required by the housing industry.

Safety Codes Act – Administration items regulations

Seals and Stamps
Complex projects
18(1) If, in the opinion of a safety codes officer, the size or complexity of a project may give rise to safety concerns, the safety codes officer may require that all plans and specifications, or any part of them, for the project's

(a) buildings,
(b) electrical systems,
(c) elevating devices,
(d) gas systems,
(e) plumbing and private sewage disposal systems, or
(f) repealed AR 49/2006 s47;
(g) fire protection systems and equipment,

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be imprinted with a seal or stamp affixed in accordance with the Engineering and Geoscience Professions Act for engineering work, or the Architects Act for architectural work.
(2) If, in the opinion of a safety codes officer, the size or complexity of a project may give rise to safety concerns, the safety codes officer may require that the construction of any or all of the project's
(a) buildings,
(b) electrical systems,
(c) elevating devices,
(d) gas systems,
(e) plumbing and private sewage disposal systems, or
(f) fire protection systems and equipment,
be reviewed during construction in accordance with the Engineering and Geoscience Professions Act or the Architects Act.
AR 16/2004 s18;49/2006;53/2016;261/2020

Unless stated otherwise, all Code references in this STANDATA are to Division B of the National Building Code - 2023 Alberta Edition

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